Islamic University of Technology
Department of Computer Science and Engineering

Lab 6: Minimum Spanning Tree

CSE 4404: Algorithms Lab
Summer 2023-24

Task A. Data Center Disaster

Time Limit: 1 second | *Memory Limit: 512 MB*

It's July 4202 and the cables connecting the data centers of a country were destroyed — due to excessive rain or, strangely enough, water dropped from helicopters.

Due to a sudden drop in foreign remittance, the government can no longer afford to repair all damaged cables. Instead, it has decided to restore connectivity by selectively repairing a subset of the cables — just enough to ensure that every data center can communicate with every other. The restoration must be done with the minimum possible cost.

Unfortunately, the engineer assigned to create the repair plan holds a BSc. in *Theatre and Performance Studies* and has long forgotten how to do math. So he has come to you for help in calculating the minimum budget for the project.

Input Format

The first line contains two integers n and m $(1 \leq n \leq 10^5,\ 1 \leq m \leq 2 \cdot 10^5)$ — the number of data centers and cables.

Then, there are m lines, each containing three integers a, b, and c $(1 \leq a, b \leq n,\ 1 \leq c \leq 10^9)$ — indicating that there is a cable between data center a and data center b, and repairing it will cost c.

All cables are bidirectional, and each connects two different data centers. There is at most one cable between any two data centers.

Output Format

Print one integer — the minimum total cost to repair enough cables so that every data center is reachable from every other, directly or indirectly.

If it is impossible to connect all data centers, print `IMPOSSIBLE`.

Examples

Sample Input	Sample Output
5 6 1 2 3 2 3 5 2 4 2 3 4 8 5 1 7 5 4 4	14

Notes

For this task, make 2 submissions with 2 different minimum spanning tree algorithms:

1. Prim's Algorithm
2. Kruskal's Algorithm

Task B. Connecting the Colony

Time Limit: 0.5 second | *Memory Limit: 512 MB*

An ant colony has n chambers arranged along a straight tunnel — numbered 1 to n from left to right. Some of these chambers are already connected to the queen's chamber via pheromone trails, while the rest are isolated.

The colony must ensure that every chamber is connected to at least one chamber that already has a pheromone trail. This is done by digging straight tunnels along the horizontal axis between chambers.

A tunnel can be built between any two chambers. Its cost is equal to the horizontal distance between their positions.

You must determine the **minimum total length of tunnels** required so that all chambers are connected to at least one with a pheromone trail.

Input Format

The first line contains a single integer T $(1 \le T \le 10)$ — the number of test cases.

The first line of each test case contains a single integer n $(1 \le n \le 10^5)$ — the number of chambers.

The second line of each test case contains a binary string of length n, where character '1' denotes that the corresponding chamber has a pheromone trail, and '0' means it does not.

The third line of each test case contains n space-separated integers $x_1, x_2, \dots, x_n$ $(1 \le x_1 < x_2 < \cdots < x_n \le 10^9)$ — the horizontal positions of the chambers.

It is guaranteed that in every test case, there is at least one chamber with a pheromone trail.

Output Format

For each test case, print a single integer: the minimum total length of tunnels needed to connect all the chambers.

Examples

Sample Input	Sample Output
2 5 10010 1 2 3 4 5 3 111 10 20 30	3 0

Marks Distribution

Task	**Marks**
Task A	60%
Task B	40%